

**CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)**

Meeting Date: 2-12-2025	Meeting No.: 2
Meeting Mode: Face-to-Face	
Project ID: 860	Project Type: Research-based
Project Title : Autonomous Driving Agent Training Using Deep Reinforcement Learning in CARLA Simulator	
Student ID : 242UC243LS	Student Name: NG JUN XUAN
Student Programme and Specialisation: Data Science	
Supervisor Name: Dr. Mohammad Shadab Khan	Co-Supervisor Name: (if applicable)
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

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1. WORK DONE

Tasks: Background Study or Literature Review

Details (in point form):

- Conduct a deeper literature review on reinforcement learning algorithms(e.g., PPO, DQN, SAC) and Transformer model (e.g. Swin Transformer, ViTransformer) used in autonomous driving .
- Formulate Research Questions (RQs) and design a detailed keyword and database search strategy.
- Execute the paper screening process, involving exclusion based on titles, abstracts, and full texts, to determine the final set of included papers.
- Develop and test a Data Extraction Form to collect key information from each paper.

2. WORK TO BE DONE

Tasks: Background Study or Literature Review / Design or Research Methodology

Details (in point form):

- Using a pre-defined data extraction form, systematically extract data from all eligible core papers (i.e., papers comparing the performance of algorithms such as PPO, SAC, and DDPG on Transformer-based policy networks).

- Based on the extracted data, we categorize and summarize the specific implementation methods, input features, and performance impact of different Transformer variants (such as Vision Transformer, Swin Transformer, etc.) in RL policy networks.

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

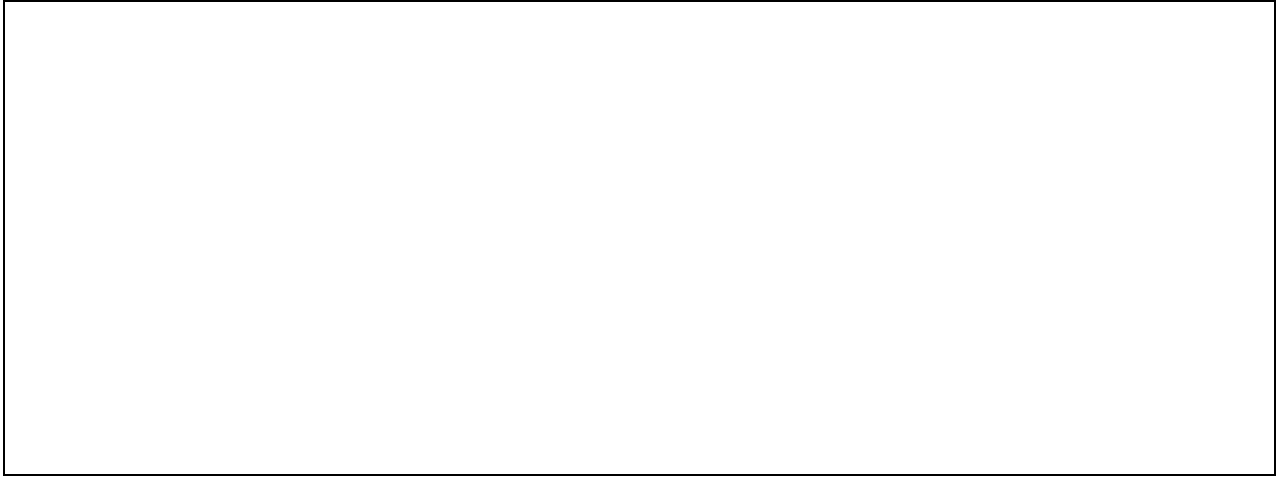
Problem:

Among the core literature selected, most papers did not directly compare the collision rates of PPO, SAC, and DDPG side-by-side under the same Transformer architecture.

Solution:

While these papers, which do not compare different algorithms within the same Transformer architecture, cannot directly answer the main research question, they can still answer other sub-research questions.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)
Satisfactory progress, research is in progress with LR.



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Supervisor's Signature

Student's Signature

Co-Supervisor's Signature
(if applicable)

Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
2. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.
3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.

4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.